

H^0 and $\tilde{T} \cdot m = 1$: Twelve Insights into Higgs Geometry

The Higgs Mechanism as Projection of T^4/Z_3 Topology

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Status Markers

- [K]** *Confirmed* — numerically verified.
- [B]** *Proven* — algebraically demonstrated.
- [S]** *Speculative* — open.
- [X]** *Experimental* — confirmed, derivation open.
- [Q]** *Source* — external primary value.

Abstract

The Higgs boson H^0 is experimentally confirmed **[X]** (CERN 2012 **[Q]**). The FFGFT statement is: the Higgs mechanism is not an independent structure but a projection of the T^4/Z_3 geometry onto the flat description. This document sets out twelve insights that together elevate this statement from **[S]** to **[B]**.

Each insight is independently derivable and individually proven. Their combined effect shows: $\tilde{T}(x, t) \cdot m(x, t) = 1$ is not postulated but forced by Einstein physics; the time field is the inverse Higgs field; apart from the motivated setting ξ the Lagrangian contains no free parameters; and the Higgs mechanism is the flat projection of the compact T^4/Z_3 topology.

1 Insight 1: $\tilde{T} \cdot m = 1$ from $E = mc^2$

Starting point: In natural units ($\hbar = c = 1$):

$$E = mc^2 = m. \quad (1)$$

The de Broglie/Compton time of a particle of mass m is

$$T_C = \frac{\hbar}{mc^2} = \frac{1}{m} \quad (\text{nat. units}). \quad (2)$$

This is not an uncertainty but the *intrinsic* timescale of the particle: the time in which the wave function completes one full phase. Hence exactly:

$$\boxed{\tilde{T}(x, t) \cdot m(x, t) = 1.} \quad (3)$$

This is a consequence of $E = mc^2$ and the de Broglie relation — not a postulate **[B]** (Docs. 003, 010, 052, 306).

Physical meaning: (3) is sharper than the energy-time uncertainty $\Delta E \cdot \Delta t \geq 1/2$ — it is an *equation*, not an inequality. Mass and intrinsic time are two readings of the same physical reality: heavy particles oscillate fast, light ones slowly **[B]** (Doc. 306).

2 Insight 2: The Time Field as Inverse Higgs Field

The Standard Model Higgs field after spontaneous symmetry breaking:

$$\Phi(x, t) = \langle \Phi \rangle + h(x, t) = v + h(x, t), \quad v \approx 246 \text{ GeV}. \quad (4)$$

The mass of a particle in the SM is $m = yv$ (Yukawa coupling y). The intrinsic time (2) is therefore:

$$\tilde{T}(x, t) = \frac{1}{m(x, t)} = \frac{1}{y(\langle \Phi \rangle + h(x, t))}. \quad (5)$$

At background level ($h = 0$):

$$\boxed{\tilde{T}_0 = \frac{1}{yv} = \frac{1}{m_0}.} \quad (6)$$

The time field is the *inverse Higgs field* **[B]** (Doc. 049).

Consequence: $\tilde{T} \cdot m = 1$ contains the vacuum expectation value implicitly: the constraint $T \cdot m = 1$ *forces* the non-vanishing of $\langle \Phi \rangle$. A vacuum with $m = 0$ would have $T \rightarrow \infty$ — a physically excluded state. The Higgs VEV is therefore not a free choice but a consequence of the reciprocity **[B]** (Docs. 306, 334).

3 Insight 3: The T0 Lagrangian

The complete T0 Lagrangian (Doc. 180) is:

$$\mathcal{L}_{T0} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell + \frac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) - \frac{1}{2}m_T^2 \Delta m^2 + \xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m. \quad (7)$$

The term $\xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m$ is the mass-proportional coupling: more massive particles couple more strongly to the time field. The coupling strength **[B]**:

$$g_T^\ell = \xi \cdot m_\ell. \quad (8)$$

In the Standard Model this role is taken by the Yukawa coupling $\mathcal{L}_Y = -y_\ell \bar{\psi}_\ell \Phi \psi_\ell$. The identification:

$$y_\ell \langle \Phi \rangle \leftrightarrow \xi \cdot m_\ell \leftrightarrow m_\ell. \quad (9)$$

For $\Delta m \rightarrow 0$ (static background) the Lagrangian (7) contains the QED part of the charged leptons with fixed masses m_ℓ as a limiting case [B] (Docs. 180, 049). This is a section, not the full SM Lagrangian: the gauge, Higgs and quark sectors are not in Eq. (7). Gravitation requires no additional field and no graviton in it (Insight 12).

4 Insight 4: Higgs EFT Consistency Check of ξ

The coupling strength $g_T^\ell = \xi \cdot m_\ell$ can be checked against the Higgs sector. The self-consistency of the Lagrangian (7) requires divergence freedom at high energies. In four spacetime dimensions the loop suppression gives [B]:

$$\frac{1}{16\pi^3} \approx 2.02 \times 10^{-3}. \quad (10)$$

Combined with the Higgs sector (Insight 2):

$$\xi_{\text{EFT}} = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} = \frac{(0.129)^2 \cdot (246.2)^2}{16\pi^3 \cdot (125.1)^2} \approx 1.30 \times 10^{-4}. \quad (11)$$

Deviation from $\xi_0 = 4/30000 = 1.333 \times 10^{-4}$: only about -2.3% (with rounded $\lambda_h = 0.129$: 1.299×10^{-4} , -2.6%). The value depends only on the rounding of the inputs: with the unrounded SM self-coupling $\lambda_h = m_h^2/(2v^2)$ one gets $\xi_{\text{EFT}} = m_h^2/(64\pi^3 v^2) = \lambda_h/(32\pi^3)$, with PDG 2024 ($m_h = 125.20$ GeV, $v = 246.22$ GeV) 1.303×10^{-4} (-2.3%); depending on the inputs the deviation lies at 2.2 – 2.6% . This is a consistency check of ξ , not an exact derivation (Doc. 385). This deviation is structurally expected (scheme-dependent SI values, fractal corrections not separable) [K] (Doc. A190).

ξ is not a free fit parameter but a motivated setting, supported by the Higgs check to within a few per cent [S].

5 Insight 5: The 10^{-4} Factor from Spacetime Dimensionality

In four spacetime dimensions the loop suppression scales as

$$\xi \sim (10^{-1})^4 = 10^{-4}. \quad (12)$$

This is not a choice — it is the consequence of $D = 4$. For $D = 3$ one would have $\xi \sim 10^{-3}$; for $D = 5$, $\xi \sim 10^{-5}$. The observed coupling strength selects $D = 4$ [B] (Doc. 180).

6 Insight 6: The $4/3$ Factor from Torus Geometry

The precise factor $4/3$ in $\xi = 4/3 \times 10^{-4}$ emerges from the harmonic structure of the T^4/Z_3 torus [B] (Docs. 009, 180):

The simplest three-dimensional structure with integer symmetry is the tetrahedron with 4 vertices and 3 edges per vertex:

$$\frac{4 \text{ vertices}}{3 \text{ edges/vertex}} = \frac{4}{3}. \quad (13)$$

This factor generates the two fundamental harmonic intervals: $3/2 \times 4/3 = 2$ (fifth $\times$ fourth = octave). The fractal torus hierarchy carries the same factor layer by layer.

The $4/3$ factor is geometrically forced, not chosen [B].

7 Insight 7: $\kappa = 7$ as the Unique Solution of the e - p - μ System

From the three fundamental mass ratios of the lepton and the proton (Doc. 180):

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^\kappa, \quad \kappa = \frac{\ln(1836.15/245)}{\ln(4/3)} = 7.000. \quad (14)$$

$\kappa = 7$ is the *only* integer solution with error $< 1\%$: $\kappa = 6$ gives 25 % error, $\kappa = 8$ gives 33 %.

The value $245 = 5 \times 7^2$ has algebraic significance in the $\text{GF}(27)^*$ framework (Doc. 338): it is the product of the Galois primes of the lepton sector. The relation $245 \cdot (4/3)^7 = 1835.43 \approx m_p/m_e$ (-0.039%) is a relation of its own; ξ does not follow from it.

8 Insight 8: The Higgs Potential from Torus Topology

The Standard Model Higgs potential:

$$V(\Phi) = \mu^2 \Phi^\dagger \Phi + \lambda(\Phi^\dagger \Phi)^2. \quad (15)$$

For $\mu^2 < 0$ the $U(1)$ symmetry breaks spontaneously; the minimum lies at $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$.

The T^4/Z_3 topology provides the same structure from geometry: the torus has a non-trivial vacuum — the winding numbers are the allowed minima. The Z_3 identification selects exactly three vacuum states (the three generations) **[B]** (Docs. 339, 346, 348).

The spontaneous symmetry breaking of the Higgs field is the *flat projection* of the topological selection of the T^4/Z_3 minimum **[B]** (Docs. A090, A190):

$$\underbrace{T^4/Z_3 \text{ topology}}_{\text{compact, [B]}} \xrightarrow{\text{projection}} \underbrace{V(\Phi), \mu^2 < 0}_{\text{flat, [X]}}. \quad (16)$$

9 Insight 9: Yukawa Couplings as Winding Numbers

In the Standard Model the Yukawa couplings y_ℓ are free parameters — 13 of them (9 charged-fermion masses and 4 CKM parameters), out of 19 free SM parameters in total. In FFGFT they are the rational winding numbers of the torus **[B]** (Docs. 006, 317, 338):

$$y_\ell \cdot v = m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v, \quad r_\ell \in \mathbb{Q}, \quad p_\ell \in \mathbb{Q}. \quad (17)$$

Concretely **[K]** ($< 1.2\%$ deviation):

Lepton	r_ℓ	p_ℓ	m_ℓ (FFGFT)
e	$4/3$	$3/2$	0.505 MeV
μ	$16/5$	1	105.1 MeV
τ	$25/9$	$2/3$	1783 MeV

The 13 free Yukawa parameters of the SM (out of 19 SM parameters in total) reduce to two numbers (r_ℓ, p_ℓ) per generation, which follow from the $\text{GF}(27)^*$ structure **[B]** (Docs. 317, 338).

10 Insight 10: Neutrino = Massless Sector in the Galois Framework

In $\text{GF}(27)^*$ the group has 26 elements, divided into orbits under the Frobenius automorphism $\phi : x \mapsto x^3$ **[B]** (Doc. 339):

- 2 fixed points $\{+1, -1\}$ — massive sector, lepton masses
- 8 three-cycles — massless sector, gluons

The three neutrino generations lie on orbit f_1 , the *massless* sector of the Galois structure **[B]** (Docs. 339, 340, 343). The neutrino carries no Galois weight. The observed neutrino masses ($m_\nu \sim \text{meV}$) are residues of the projection, not Galois masses **[K]**:

$$m_{\nu_i} = \text{projection residue}, \quad \text{Galois weight}(f_1) = 0. \quad (18)$$

Consequence for the Higgs: The neutrino has no Yukawa coupling in the Galois framework — it does not couple to the Higgs field. The SM Yukawa $y_\nu \bar{\psi}_\nu \Phi \psi_\nu$ is in FFGFT not a fundamental term but a fitting parameter without geometric basis **[B]** (Docs. 343, 340).

ν_3 is a Dirac neutrino (orbit 4 is not self-inverse) **[B]**; ν_1, ν_2 are Majorana (orbits 1, 2 are self-inverse) **[B]** (Docs. 340, 343).

11 Insight 11: Higgs = Vacuum, Not an Independent Particle

In the Standard Model the Higgs boson is a particle — the excitation around the vacuum $|\Phi|^2 = v^2/2$. In FFGFT this picture is a projection: the vacuum *is* the time field $\tilde{T}(x, t)$ at background level, and H^0 is the observable of this projection structure.

The analogy to the graviton search is exact:

	Graviton	Higgs H^0
SM/GR view	Carrier of gravitation	Carrier of mass
FFGFT view	G emergent from ξ (I12)	$\tilde{T} \cdot m = 1$ structural (I1)
What is measured	Gravitational waves (LIGO)	Fluctuation around vacuum (CERN)
What it is <i>not</i>	Fundamental particle	Independent excitation
Status	G is derived [K]	H^0 is projection residue [B]

Physically: $\tilde{T}(x, t) = 1/(\langle \Phi \rangle + h(x, t))$ (Insight 2). In the vacuum ($h = 0$) the time field is constant: no particle. The fluctuation $h(x, t)$ is what CERN measures **[X]**. In FFGFT h is not an independent excitation but the visible component of the time field fluctuations $\Delta m(x, t)$ (Lagrangian (7)) **[B]**.

12 Insight 12: G Is Not Fundamental

The gravitational constant follows from ξ and the characteristic mass **[K]** (Docs. 012, 180):

$$G = \frac{\xi^2}{4 m_e} \quad (\text{T0 natural units}), \quad (19)$$

with SI corrections $C_{\text{dim}}, C_{\text{conv}}, K_{\text{frak}}$ (Doc. 012): with $K_{\text{frak}} = 0.986$, $G_{\text{SI}} \approx 6.6745 \times 10^{-11} \text{ N m}^2/\text{kg}^2$, deviation $+0.003\%$; with $K_{\text{frak}} = 1 - 100\xi = 74/75$ the result is 6.679×10^{-11} ($+0.07\%$). The form and order of magnitude of G follow from ξ and the single anchor **[K]**. The close agreement with the CODATA value arises only with the small residual of the one-anchor chain contained in the numerical value of C_{conv} (Doc. 383) and does not count as an independent precision confirmation.

G is therefore not a fundamental parameter but an *emergent ratio* from ξ and m_e **[K]** (Doc. 012).

The three gauge hierarchies of the SM thus have different roles in FFGFT:

Interaction	SM status	FFGFT status	Doc.
Strong (8 gluons)	fundamental	$4 \times Z_{13}$ orbits [B]	339
Electroweak ($W^\pm, Z^0, \gamma$)	fundamental	$\text{Gal}(GF(9)/GF(3)) = Z_2$ [K]	336
Higgs H^0	particle	projection of vacuum [B]	this doc.
Gravitation (G)	fundamental	emergent from ξ [K]	012
Graviton	sought	perturbative residue, not a particle [B]	this doc.

13 Overview: From [X] to [B]

	Insight	Status	Source
1	$\tilde{T} \cdot m = 1$ from $E = mc^2$ + de Broglie	[B]	Docs. 003, 010, 052, 306
2	Time field = inverse Higgs field	[B]	Doc. 049
3	T0 Lagrangian contains the lepton QED part of the SM as limit	[B]	Docs. 180, 049
4	Higgs EFT consistency check of ξ (about 2.3 %)	[K]	Docs. 180, A190
5	10^{-4} factor from $D = 4$ spacetime	[B]	Doc. 180
6	$4/3$ factor from torus geometry	[B]	Docs. 009, 180
7	$\kappa = 7$ unique solution of e - p - μ system	[K]	Docs. 180, 338
8	Higgs potential = projection of T^4/Z_3 topology	[B]	Docs. A090, A190
9	Yukawa couplings = winding numbers $GF(27)^*$	[K]	Docs. 317, 338
10	Neutrino = massless Galois sector, no Yukawa	[B]	Docs. 339, 340, 343
11	H^0 = projection residue of vacuum, not a particle	[B]	Docs. 049, A090, A190
12	G emergent from ξ ; graviton not a particle	[K]	Docs. 012, 180
	H^0 experimentally confirmed	[X]	CERN 2012 [Q]

Conclusion: H^0 is experimentally confirmed **[X]**. The twelve insights together give:

- The Higgs mechanism is the flat projection of the T^4/Z_3 geometry **[B]**.
- $\tilde{T} \cdot m = 1$ is forced from $E = mc^2$, not postulated **[B]**.
- H^0 is the perturbative degree of freedom of the vacuum **[B]**.
- The neutrino carries no Galois weight; its mass is a projection residue **[B]**.
- G is emergent from ξ ; the graviton is not a fundamental particle **[K]**.
- Coupling strengths g_s, g_w derived from ξ **[K]**; e depends via α on the anchor E_0 .
- SSB $SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$: $\langle \Phi \rangle \neq 0$ derived **[B]**; the neutrality of the vacuum from the Galois structure is open **[S]** (Doc. 355, chain steps 4 and 6).
- CKM/PMNS angle values: remaining structural **[S]**.

Clarification of the gauge sector status after Doc. A192:

The gauge group structure is not open — it is topologically forced **[B]** (Doc. A192):

- $U(1)_Y$ from flux quantisation on non-contractible torus loop **[B]**
- $SU(3)_C$ from linking number of three $\mathbb{Z}_3$ flux tubes; $6 + 2 = 8$ generators topologically proven **[B]**
- $SU(2)_L$ from chiral projection of the spin- $\frac{1}{2}$ sector (A095) **[B]**

The complete gauge Lagrangian $\mathcal{L}_{\text{gauge}}$ follows without free choice **[B]**. What the SM particle diagram (Doc. 356) displays (quantum numbers Q, Y , colour charge, chirality, generations) is established as **[B]** and **[K]**, except for the hypercharge of the charged leptons, which does not follow from Theorem A of Doc. 347 **[S]**; the detailed certificate status of each quantum number is given in Doc. 357. The coupling strengths g_s, g_w, e do not appear in the classification table at all — they belong to Lagrangian dynamics, not to group classification.

14 Resolution of the Three Open Points

[S1] Coupling constants g_s, g_w, e from ξ — closed for g_s, g_w **[K]**

The strong and the weak coupling strengths are derived from ξ ; the electromagnetic one depends on the anchor E_0 :

Electromagnetic coupling e :

$$\alpha_{\text{EM}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2, \quad e = \sqrt{4\pi\alpha_{\text{EM}}} \quad (20)$$

with $E_0 = \sqrt{m_e m_\mu}$ **[K]** (Docs. A130, 011). The agreement of $\alpha = \xi E_0^2$ with the measured value is a property of the anchor E_0 , which provides the only connection to the SI; it is not counted as an independent prediction.

Strong coupling g_s :

$$\alpha_s(m_\tau) = D \cdot \xi^{1/(D+1)} = 3 \cdot \xi^{1/4} = 0.3224, \quad g_s = \sqrt{4\pi\alpha_s} \quad (21)$$

Deviation from experiment: -2.3% **[K]** (Doc. 160).

Weak coupling g_w :

$$\sin^2 \theta_W|_{\text{GUT}} = \frac{3}{8} \xrightarrow{\text{RGE running}} \sin^2 \theta_W(M_Z) = 0.2308 \quad (22)$$

GUT value from trace formula of five $T^4/\mathbb{Z}_3$ orbifold states **[B]** (Doc. 321); RGE running gives -0.19% from PDG **[K]** (Doc. 323).

[S2] SSB $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$ — partly closed **[B]/[S]**

Three building blocks, all in the corpus:

(a) $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$ **[B]**.

(b) Neutrality of the vacuum, $Q_{\text{vac}} = 0$ — open **[S]**: $Q = 0 \iff \text{QR mod } 13$ (Doc. 346 Theorem A), but the topologically trivial vacuum element ($k = 0$) has Legendre symbol 0 and is neither QR nor NQR; that it is QR is not established (Doc. 355, chain step 4). Likewise the hypercharge $Y = +1$ of the Higgs doublet does not follow from Doc. 347 Theorem A, which gives $Y = -1$ for QR (Doc. 355, chain step 6).

(c) $Q = 0$ vacuum leaves $U(1)_{\text{EM}}$ invariant: $e^{i\alpha Q}|\text{vac}\rangle = |\text{vac}\rangle$ **[B]**.

Together: (a) and (c) hold **[B]**; that the Galois structure forces $Q_{\text{vac}} = 0$ is open **[S]** (Doc. 355, chain steps 4 and 6).

[S3] CKM/PMNS angles — remains open **[S]**

Structure of mixing matrices proven **[B]** (Doc. 348 Theorems B,C). Angle values require further algebraic work (Doc. 348 Theorem D). Besides the neutrality of the vacuum ([S2]) this is the remaining structural **[S]** in the Higgs/gauge complex.

15 What This Document Does *Not* Claim

- Not: the Higgs mechanism is wrong — it is correct and confirmed **[X]**.
- Not: FFGFT needs no Higgs field — it appears as a projection.
- Not: the EFT agreement (11) is a proof — it is a consistency finding at about 2.3 % (with rounded $\lambda_h = 0.129$: 2.6 %) **[K]**.
- Not: Yukawa couplings fully derived — rational winding numbers agree at $< 1.2 \%$, but K-041-a/b/c remain open (Doc. 190 register).

16 Verification Script

- Numerical verification of all twelve insights (EFT agreement, $\kappa = 7$ uniqueness, Yukawa values, 10^{-4} factor): `python/Dok354_Skripte/pruef_354_higgs_tm.py`

17 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.